

B.Tech Project (CP303)



Title: Committee Selection in Temporal Settings

Group: 15

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Contents

- ➡ **MultiWinner Voting**
- ➡ **Introduction to Temporal Settings**
- ➡ **Our Approach**
- ➡ **Problem Settings**
- ➡ **Computational Complexity**
- ➡ **Welfare Function**
- ➡ **Welfare-Fairness Tradeoff**
- ➡ **ILP Formulation**
 - ➡ **Key Constraints**
- ➡ **Implementation**
- ➡ **Future Proof**
- ➡ **References**

Multiwinner Voting

Multiwinner voting is the process of selecting a group of winners (a committee) instead of a single candidate.

Some classical multiwinner voting rules:

- **AV(Approval Voting)**: Pick k candidates with the most approvals.
- **PAV(Proportional Approval Voting)**: Ensures proportional representation.
- **CCAV(Chamberlin-Courant)**: Each voter gets a representative.
- **SAV(Satisfaction Approval Voting)**: Maximize total satisfaction.
- **NAV/NSAV**: Approvals minus disapprovals

Introduction to Temporal Settings

Classical multiwinner voting selects one committee for one election.

In many real tasks, preferences change over time, and we need to elect a sequence of committees, one for each time step (day/event/year).

Examples include:

- Rotating task assignment teams
- Hiring or evaluation panels
- Scheduling service committees

In such cases, we must elect a sequence of committees instead of one committee.

where $C_1, C_2, \dots, C_T$

- C_t = committee selected at round t
- T = total number of rounds

Key challenge:

Voter preferences and approvals may change across rounds.

Research Gap

Research Gap

Existing work on temporal committee selection focuses on successive committee algorithms, fairness notions, and computational complexity. Welfare-based optimization in temporal committee settings has received little attention, unlike its single-winner counterpart studied by Elkind et al ,Tzeh Y and Nicholas T in Temporal Elections: Welfare, Strategyproofness, and Proportionality.

Gap

While they establish welfare maximization frameworks for temporal single-winner settings, no analogous formulation exists for temporal committee selection. In particular, the welfare-fairness tradeoff in committee sequences, and its computation via ILP are still limited.

Our Approach

To address the temporal committee selection problem, we follow a structured framework:

TEMPORAL VOTING MODEL:

We consider a sequence of elections over time where, at each timestep, a committee of fixed size k is selected based on voters' approval preferences.

WELFARE FUNCTIONS:

We evaluate outcomes using two social welfare notions:

- **Utilitarian welfare:** total number of voter representation events.
- **Egalitarian welfare:** representation of the least satisfied voter.

WELFARE-FAIRNESS TRADEOFF:

We study the balance between maximizing overall welfare while ensuring that no voter remains persistently under-represented.

OPTIMIZATION VIA ILP:

We design an Integer Linear Programming formulation that computes an optimal sequence of committees maximizing welfare while incorporating fairness considerations.

Problem Settings

Given:

- **Agents:** $N = \{1, \dots, n\}$
- **Candidates:** P
- **Timesteps:** $t = 1, \dots, \ell$
- **Committee size:** k

At each timestep t :

$$C_t \subseteq P, |C_t| = k$$

Each voter has approval set $S_{i,t}$.

A voter is represented at time t if:

$$C_t \cap S_{i,t} \neq \emptyset$$

Our goal is to compute a sequence of committees that maximizes utilitarian welfare aggregated over all timesteps while satisfying egalitarian fairness constraints that guarantee each agent a minimum level of representation over the time horizon.

Computational Complexity

NP-Hardness

Computing an optimal temporal committee that maximizes welfare while ensuring fairness is computationally difficult.

We show this using a reduction from the **Maximum Set Coverage problem**.

We map the Set Coverage instance to the voting model:

- **Elements** $\rightarrow$ Voters
- **Sets** $\rightarrow$ Candidates
- **Set coverage** $\rightarrow$ Voter representation

Selecting k candidates that maximize represented voters is equivalent to selecting k sets covering the maximum elements.

Welfare Function

We consider two classical social welfare objectives.

Utilitarian Welfare:

Keep committees stable over time.

$$\text{UTIL}(\mathbf{C}) = \sum_{i \in N} u_i(\mathbf{C}),$$

$$u_i(\mathbf{C}) = |\{t \in [\ell] \mid C_t \cap S_{i,t} \neq \emptyset\}|$$

Egalitarian Welfare:

The egalitarian welfare of an outcome $\mathbf{C}$ is defined as

$$\text{EGAL}(\mathbf{C}) = \min_{i \in N} u_i(\mathbf{C}),$$

Welfare-Fairness Tradeoff

01

AGENT REPRESENTATION

For a committee sequence $\mathbf{C}=(\mathbf{C}_1,\dots,\mathbf{C}_\ell)$

The representation of agent i is:

$$u_i(\mathbf{C}) = |\{t \in [\ell] \mid \mathbf{C}_t \cap S_{i,t} \neq \emptyset\}|$$

This counts how many rounds agent i is represented.

02

MINIMUM REPRESENTATION

To measure fairness we define:

$$\lambda(\mathbf{C}) = \min_{i \in N} u_i(\mathbf{C}).$$

This represents the satisfaction of the least represented agent.

03

TRADEOFF PRINCIPLE

1.Fairness Constraint:

A committee sequence is F -fair if

$$\lambda(\mathbf{C}) \geq F$$

This ensures every agent receives a minimum level of representation.

2.Optimization principle

Among all F -fair committee sequences we apply a lexicographic objective:

- Maximize utilitarian welfare
- Break ties by maximizing minimum representation

ILP Formulation

DECISION VARIABLES:

Candidate Selection

$$x_{p,t} \in \{0, 1\}$$

Agent representation

$$y_{i,t} \in \{0, 1\}$$

Cumulative representation

$$z_{i,t} \geq 0$$

Fairness variable

$$\lambda$$

Objective Function:

$$\max \sum_{i \in N} \sum_{t=1}^{\ell} y_{i,t} + \varepsilon \lambda,$$

Optimization Objective:

$$\varepsilon < \frac{1}{|N| \cdot \ell + 1},$$

Where:

- first term = total welfare
- second term = minimum cumulative representation

Key Constraints

Committee Size

$$\sum_{p \in P} x_{p,t} = k \quad \forall t \in [\ell].$$

Representation

Voter represented if approved
candidate selected.

Fairness Guarantee

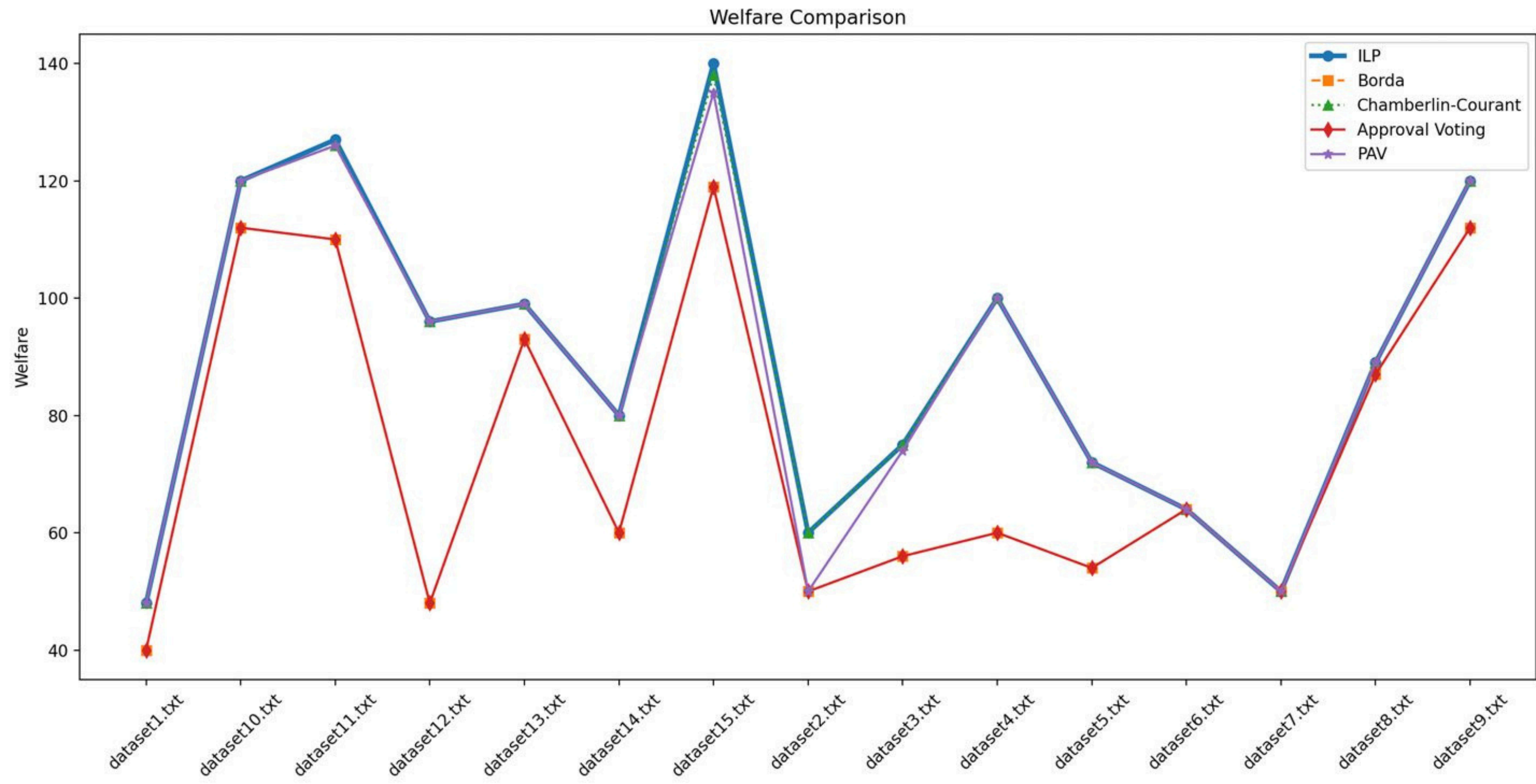
$$\lambda \leq z_{i,\ell} \quad \forall i \in N.$$

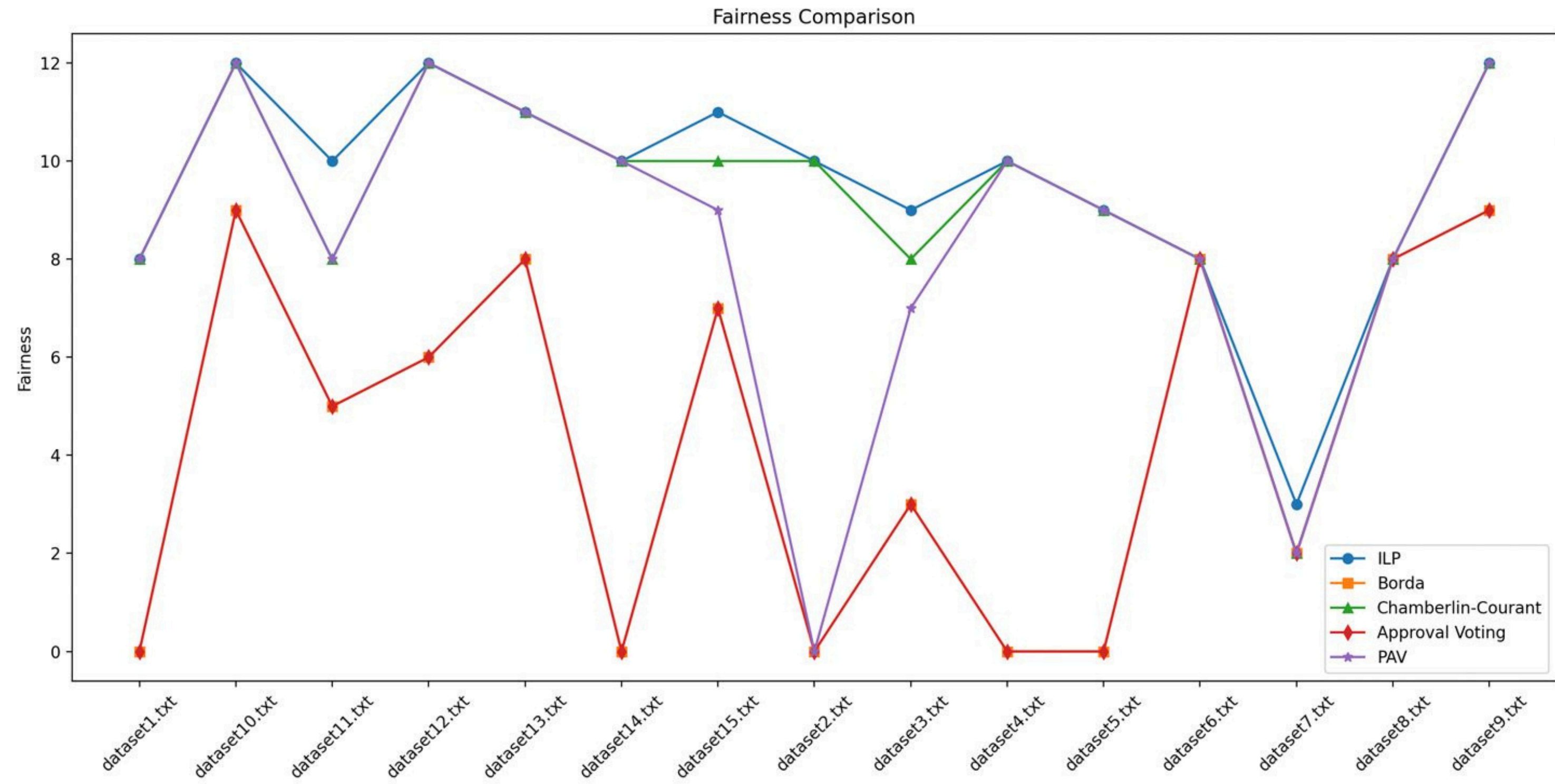
Cumulative Representation

$$\begin{aligned} z_{i,1} &= y_{i,1} & \forall i \in N, \\ z_{i,t} &= z_{i,t-1} + y_{i,t} & \forall i \in N, t = 2, \dots, \ell. \end{aligned}$$

Implementation

- ➡ Implemented the proposed ILP formulation in Python using **PuLP** to compute the optimal sequence of temporal committees.
- ➡ Implemented baseline voting rules for comparison:
 - Approval Voting (AV)
 - Borda
 - Chamberlin–Courant (CC)
 - Proportional Approval Voting (PAV)
- ➡ Experimental comparison is performed between the ILP solution and these classical voting rules using welfare and fairness metrics.





Future Work

- Strategyproofness analysis:

Study whether agents can manipulate the outcome by misreporting their approvals.

- Bound on F:

Derive theoretical bounds on the minimum fairness guarantee achievable in temporal settings.

- Larger real-world datasets:

Evaluate the model on larger and more realistic temporal voting datasets.

References

Research Papers:

- Electing Successive Committees: Complexity and Algorithms
- Efficient Algorithms for Electing Successive Committees
- Multiwinner Voting with Restricted Admissible Sets: Complexity and Strategyproofness
- Temporal Fairness in Multiwinner Voting
- Strengthening Proportionality in Temporal Voting
- Temporal Elections: Welfare, Strategyproofness, and Proportionality

Q & A